

Statistics, Counting & Probability



The fundamental counting principle

- 1 A restaurant menu has 4 appetizers, 6 main courses, and 3 desserts. How many different 3-course meals are possible?
 - 2 A license plate consists of 3 letters followed by 4 digits. How many different license plates are possible, assuming repetition is allowed?
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Permutations

- 3 In how many ways can 5 different books be arranged on a shelf?
 - 4 In how many ways can a president, vice president, and secretary be chosen from a club of 10 members?
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Combinations

- 5 In how many ways can a hand of 5 cards be dealt from a standard 52-card deck?
 - 6 A committee of 4 is chosen from 6 men and 5 women. In how many ways can the committee consist of exactly 2 men and 2 women?
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Probability with combinations

- 7 A bag contains 5 red and 7 blue marbles. If 3 marbles are drawn at random without replacement, find the probability all 3 are red.
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The binomial probability formula

- 8 A fair coin is flipped 5 times. Find the probability of getting exactly 3 heads, using the binomial probability formula $P(k) = \binom{n}{k} p^k (1 - p)^{n - k}$.
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Expected value and variance

- 9 A game gives a payout of 10 dollars with probability 0.3, 5 dollars with probability 0.4, and 0 dollars with probability 0.3. Find the expected payout.

Normal distribution and z-scores

- 10 A data set is normally distributed with mean 70 and standard deviation 8. Find the z-score for a data value of 86.
- 11 Using the empirical rule (68-95-99.7 rule), estimate the percentage of data within 2 standard deviations of the mean.

A multi-part probability and counting problem

- 12 This question has two parts. A class has 8 boys and 7 girls. A group of 4 students is selected at random to represent the class. (a) Find the total number of ways to select the group of 4 from the 15 students. (b) Find the probability that the group consists of exactly 2 boys and 2 girls.

Permutations with repeated elements

- 13 Find the number of distinct arrangements of the letters in the word "BANANA".

Conditional probability with counting

- 14 A committee of 3 is selected from 5 men and 4 women. Given that at least one woman is selected, find the probability that exactly 2 women are selected.

Percentiles and standardized scores

- 15 A student scores at the 85th percentile on a standardized test. Explain what this means in terms of the rest of the test-taking population.

Mutually exclusive vs. overlapping events

- 16 A die is rolled once. Find $P(\text{rolling an even number or a 5})$, explaining why the addition rule simplifies since these events are mutually exclusive.
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A multi-part counting and probability problem with restrictions

- 17 This question has three parts. A password consists of 4 characters chosen from the 10 digits 0-9, and repetition is not allowed. (a) Find the total number of possible passwords. (b) Find the number of passwords where the first character is not 0 (a common restriction to avoid leading zeros). (c) Find the probability that a randomly generated valid password (from part b) starts with an even digit.
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Multiple choice — easy

- 18 A store offers 8 toppings, 2 drink flavors, and 5 wallpaper patterns. How many different combinations of one item from each category are possible?
- a. 88
 - b. 80
 - c. 18
 - d. 21
- 19 A store offers 2 crust types, 7 pant styles, and 3 wallpaper patterns. How many different combinations of one item from each category are possible?
- a. 42
 - b. 17
 - c. 23
 - d. 44
- 20 A store offers 9 font styles, 2 shoe types, and 9 crust types. How many different combinations of one item from each category are possible?
- a. 162
 - b. 165
 - c. 171
 - d. 27

- 21** A store offers 3 drink flavors, 2 appetizers, and 2 font styles. How many different combinations of one item from each category are possible?
- a. 12
 - b. 7
 - c. 15
 - d. 8
- 22** A store offers 6 book genres, 6 appetizers, and 4 shoe types. How many different combinations of one item from each category are possible?
- a. 150
 - b. 30
 - c. 40
 - d. 144
- 23** A store offers 9 main courses, 6 wallpaper patterns, and 7 book genres. How many different combinations of one item from each category are possible?
- a. 61
 - b. 387
 - c. 378
 - d. 51
- 24** A store offers 6 book genres, 4 toppings, and 6 shirt colors. How many different combinations of one item from each category are possible?
- a. 144
 - b. 150
 - c. 30
 - d. 147

- 25** A store offers 5 wallpaper patterns, 5 drink flavors, and 5 book genres. How many different combinations of one item from each category are possible?
- a. 30
 - b. 128
 - c. 130
 - d. 125
- 26** A store offers 7 shoe types, 7 shirt colors, and 2 desserts. How many different combinations of one item from each category are possible?
- a. 51
 - b. 98
 - c. 21
 - d. 105
- 27** A store offers 4 book genres, 2 main courses, and 6 wallpaper patterns. How many different combinations of one item from each category are possible?
- a. 52
 - b. 16
 - c. 48
 - d. 14
- 28** A store offers 4 main courses, 2 desserts, and 8 toppings. How many different combinations of one item from each category are possible?
- a. 68
 - b. 16
 - c. 20
 - d. 64

- 29** A store offers 7 crust types, 3 toppings, and 7 appetizers. How many different combinations of one item from each category are possible?
- a. 150
 - b. 28
 - c. 154
 - d. 147
- 30** A store offers 4 shoe types, 2 desserts, and 5 pant styles. How many different combinations of one item from each category are possible?
- a. 44
 - b. 13
 - c. 40
 - d. 14
- 31** A store offers 6 desserts, 6 main courses, and 9 pant styles. How many different combinations of one item from each category are possible?
- a. 60
 - b. 324
 - c. 330
 - d. 45
- 32** A store offers 7 crust types, 2 toppings, and 5 appetizers. How many different combinations of one item from each category are possible?
- a. 77
 - b. 70
 - c. 19
 - d. 17

Multiple choice — medium

- 33** In how many ways can a committee of 5 people be chosen from a group of 15 people?
- a. 3003
 - b. 3008
 - c. 360360
 - d. 5005
- 34** In how many ways can a committee of 3 people be chosen from a group of 16 people?
- a. 3360
 - b. 560
 - c. 1820
 - d. 563
- 35** In how many ways can a committee of 5 people be chosen from a group of 10 people?
- a. 30240
 - b. 252
 - c. 257
 - d. 210
- 36** In how many ways can a committee of 3 people be chosen from a group of 12 people?
- a. 1320
 - b. 220
 - c. 223
 - d. 495

- 37 In how many ways can a committee of 8 people be chosen from a group of 18 people?
- a. 43766
 - b. 1764322560
 - c. 48620
 - d. 43758
- 38 In how many ways can a committee of 2 people be chosen from a group of 12 people?
- a. 220
 - b. 66
 - c. 68
 - d. 132
- 39 In how many ways can a committee of 12 people be chosen from a group of 15 people?
- a. 105
 - b. 455
 - c. 467
 - d. 217945728000
- 40 In how many ways can a committee of 6 people be chosen from a group of 15 people?
- a. 5011
 - b. 6435
 - c. 5005
 - d. 3603600

- 41 In how many ways can a committee of 8 people be chosen from a group of 14 people?
- a. 3011
 - b. 3003
 - c. 2002
 - d. 121080960
- 42 In how many ways can a committee of 5 people be chosen from a group of 8 people?
- a. 56
 - b. 61
 - c. 6720
 - d. 28
- 43 In how many ways can a committee of 3 people be chosen from a group of 13 people?
- a. 286
 - b. 289
 - c. 715
 - d. 1716
- 44 In how many ways can a committee of 3 people be chosen from a group of 8 people?
- a. 56
 - b. 59
 - c. 336
 - d. 70

- 45 In how many ways can a committee of 6 people be chosen from a group of 8 people?
- a. 8
 - b. 28
 - c. 34
 - d. 20160
- 46 In how many ways can a committee of 3 people be chosen from a group of 15 people?
- a. 1365
 - b. 2730
 - c. 458
 - d. 455
- 47 In how many ways can a committee of 4 people be chosen from a group of 9 people?
- a. 130
 - b. 126
 - c. 3024
 - d. 135

Multiple choice — hard

- 48 A bag contains 7 red and 6 blue marbles. If 3 marbles are drawn at random without replacement, find the probability all 3 are red.
- a. $\frac{10}{143}$
 - b. $\frac{7}{13}$
 - c. $\frac{35}{286}$
 - d. $\frac{35}{13}$

- 49** A bag contains 4 red and 5 blue marbles. If 4 marbles are drawn at random without replacement, find the probability all 4 are red.
- a. $5/126$
 - b. $4/9$
 - c. $1/126$
 - d. $1/9$
- 50** A bag contains 5 red and 5 blue marbles. If 4 marbles are drawn at random without replacement, find the probability all 4 are red.
- a. $1/14$
 - b. $1/2$
 - c. $1/21$
 - d. $1/42$
- 51** A bag contains 8 red and 5 blue marbles. If 3 marbles are drawn at random without replacement, find the probability all 3 are red.
- a. $56/13$
 - b. $28/143$
 - c. $8/13$
 - d. $5/143$
- 52** A bag contains 6 red and 6 blue marbles. If 3 marbles are drawn at random without replacement, find the probability all 3 are red.
- a. $1/11$
 - b. $5/3$
 - c. $2/11$
 - d. $1/2$

- 53 A fair coin is flipped 4 times. Find the probability of getting exactly 2 heads, using the binomial probability formula $P(k) = \binom{n}{k}p^k(1-p)^{n-k}$.
- a. $1/2$
 - b. $3/8$
 - c. $3/4$
 - d. $5/8$
- 54 A fair coin is flipped 8 times. Find the probability of getting exactly 5 heads, using the binomial probability formula $P(k) = \binom{n}{k}p^k(1-p)^{n-k}$.
- a. $5/16$
 - b. $1/4$
 - c. $7/32$
 - d. $9/32$
- 55 A fair coin is flipped 4 times. Find the probability of getting exactly 1 heads, using the binomial probability formula $P(k) = \binom{n}{k}p^k(1-p)^{n-k}$.
- a. 1
 - b. $1/4$
 - c. $3/4$
 - d. $1/2$
- 56 A fair coin is flipped 7 times. Find the probability of getting exactly 1 heads, using the binomial probability formula $P(k) = \binom{n}{k}p^k(1-p)^{n-k}$.
- a. $1/16$
 - b. 1
 - c. $9/128$
 - d. $7/128$

- 57 A fair coin is flipped 5 times. Find the probability of getting exactly 4 heads, using the binomial probability formula $P(k) = \binom{n}{k}p^k(1-p)^{n-k}$.
- a. 1
 - b. 5/32
 - c. 7/32
 - d. 3/16
- 58 Find the number of distinct arrangements of the letters in the word "PIZZA".
- a. 60
 - b. 120
 - c. 58
 - d. 63
- 59 Find the number of distinct arrangements of the letters in the word "APPLE".
- a. 60
 - b. 58
 - c. 63
 - d. 120
- 60 Find the number of distinct arrangements of the letters in the word "TENNESSEE".
- a. 3780
 - b. 362880
 - c. 3684
 - d. 7560

61 Find the number of distinct arrangements of the letters in the word "COMMITTEE".

- a. 45360
- b. 90720
- c. 45352
- d. 362880

62 Find the number of distinct arrangements of the letters in the word "BANANA".

- a. 120
- b. 48
- c. 720
- d. 60